# CVP Confinement Layer for Fusion: Invariant-Driven Control, TEVR Proof Bundles, and Canonical Equation Governance

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***Abstract*—Controlled fusion remains constrained by turbulence-driven transport, MHD instabilities, heat-exhaust limits, and the operational reality that expensive shots demand repeatable progress. This paper presents the *CVP Confinement Layer* as a device-agnostic stability and transport control overlay: invariant-driven control metrics, strict local recovery constraints, and TEVR proof bundles (deterministic replay) to validate claimed improvements. We also lock canonical equation governance to prevent notation drift: $c$ remains the local invariant causal speed, while $c_{\mathrm{eff}}$ is an effective propagation variable used for curvature-conditioned control and transport modeling under a gravitational-geometric state $G_{\mathrm{geom}}$. A staged validation plan (Phase A → Phase C) and acceptance gates are provided to support partner integration and auditability.**

***Index Terms*—fusion control, plasma stability, transport suppression, robust control, reinforcement learning, diagnostics, reproducibility, TEVR, Curvature Variable Physics**

## I. Canonical Equation Panel (Locked)

**Boundary condition.** In this document, $c$ remains locally invariant. $c_{\mathrm{eff}}$ is an *effective* propagation variable used for curvature-conditioned control and transport modeling; it does not assert a global change to local relativity.

**Locked forms.**

$$c \;=\; \text{local invariant causal speed} \tag{1}$$

$$c_{\mathrm{eff}} \;=\; f(K, R, \nabla K, \partial_t K) \tag{2}$$

$$D(c_{\mathrm{eff}}, h, G_{\mathrm{geom}}) \;=\; \left.\frac{\delta c_{\mathrm{eff}}}{\delta G_{\mathrm{geom}}}\right|_h \tag{3}$$

$$D(c_{\mathrm{eff}}, h, G_{\mathrm{geom}}) \;=\; \left.\frac{\partial c_{\mathrm{eff}}}{\partial G_{\mathrm{geom}}}\right|_h \tag{4}$$

$$\Delta_G c_{\mathrm{eff}} \;=\; c_{\mathrm{eff}} - c \tag{5}$$

$$T_{\mu\nu} \;\to\; G_{\mathrm{geom}} \;\to\; c_{\mathrm{eff}} \tag{6}$$

**Notation governance.** Use $G_{\mathrm{geom}}$ (or $\mathcal{G}$) for gravitational geometry; avoid plain $G$ for the Dillon Operator.

## II. Motivation

Fusion progress increasingly depends on control: keeping the system inside a narrow corridor where it is stable, performant, and safe for plasma-facing components. Modern efforts in learned plasma control, fast differentiable simulation, and virtual diagnostics demonstrate the direction of travel. The CVP Confinement Layer is designed to complement these efforts with a governance-first control overlay: stable invariant state, hard constraints, local recovery, and deterministic replay.

## III. CVP Confinement Layer Concept

The Confinement Layer is a structured control architecture built on four commitments:

1) **Invariant-driven state** $I(t)$: robust summaries of machine+plasma configuration that reduce sensitivity to sensor drift and day-to-day changes.
2) **Strict local recovery**: actuator-off recovery and weak-perturbation recovery must hold within declared tolerance $\varepsilon$.
3) **Actuation families**: waveform/geometry families parameterized by $\theta$ (e.g., bias waveforms, auxiliary modulation) subject to hard limits.
4) **TEVR proof bundles**: deterministic provenance from raw traces to metrics and figures.

## IV. Primary Metrics

We evaluate control authority using two primary metrics.

### A. Band-limited fluctuation power

Compute a power spectral density (PSD) on a selected signal (probe, photodiode proxy, or equivalent). Define

$$\Pi(\theta, I) \;=\; \int_{f_1}^{f_2} \mathrm{PSD}(f;\theta, I)\, df. \tag{7}$$

A successful actuator reduces $\Pi$ at fixed baseline setpoints.

### B. Afterglow decay proxy

For afterglow tests, fit

$$I_{\mathrm{em}}(t) \approx I_0\, e^{-t/\tau(\theta, I)} + b \tag{8}$$

and compare $\tau$ baseline vs. CVP-on under matched conditions.

TABLE I
SYMBOLS (CANONICAL, COMPACT)

| Symbol | Meaning |
|---|---|
| $c$ | Local invariant causal speed (boundary condition). |
| $c_{\text{eff}}$ | Effective propagation variable for control/transport modeling (Eq. 2). |
| $G_{\text{geom}}$ | Gravitational-geometric state (notation governance). |
| $I(t)$ | Invariant-driven control metric vector (device-robust state). |
| $\theta$ | Actuator waveform parameters (family index + continuous params). |
| $\Pi$ | Integrated band-limited fluctuation power. |
| $\tau$ | Afterglow decay proxy for confinement. |
| $\varepsilon$ | Declared replay tolerance / recovery tolerance (TEVR). |

TABLE II
STAGE-GATED ACCEPTANCE CRITERIA (TEVR)

| Gate | Environment | Acceptance criteria |
|---|---|---|
| A | Phase A testbed | $\Pi$ reduction and $\tau$ improvement beyond $\varepsilon$; actuator-off and weak-perturbation recovery pass; deterministic replay succeeds. |
| B | Neutron-visible | Improved neutrons/J with replication; no constraint violations; TEVR evidence progression (E3→E4). |
| C | Partner device | Improved stability threshold or confinement KPI; multi-day repeatability; TEVR E4/E5 targets. |

## V. Symbol Table

## VI. TEVR Acceptance Gates

## VII. Discussion: Why an Overlay

Fusion needs not only better performance, but *repeatable convergence*. An overlay approach explicitly binds learning-based controllers to safety constraints and recovery obligations, then packages results into replayable evidence. This reduces the chance of mistaking drift, tuning artifacts, or one-off operating points for real progress.

## VIII. Conclusion

The CVP Confinement Layer is presented as a governance-first stability and transport control overlay: invariant state, strict local recovery, hard constraints, and TEVR proof bundles. The canonical equation panel locks notation drift ($c$ vs. $c_{\text{eff}}$; $G_{\text{geom}}$ governance) while maintaining local invariance of $c$. The next step is Phase A validation in a diagnostic-friendly testbed, followed by staged acceptance gates to neutron-visible and partner-device environments.

## References


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